Operating System Structures

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Outline

- What services are provided by the OS for
  - Users
  - Processes
  - Other systems
- Different operating system structures
- Operating systems: installing and booting

OS Services

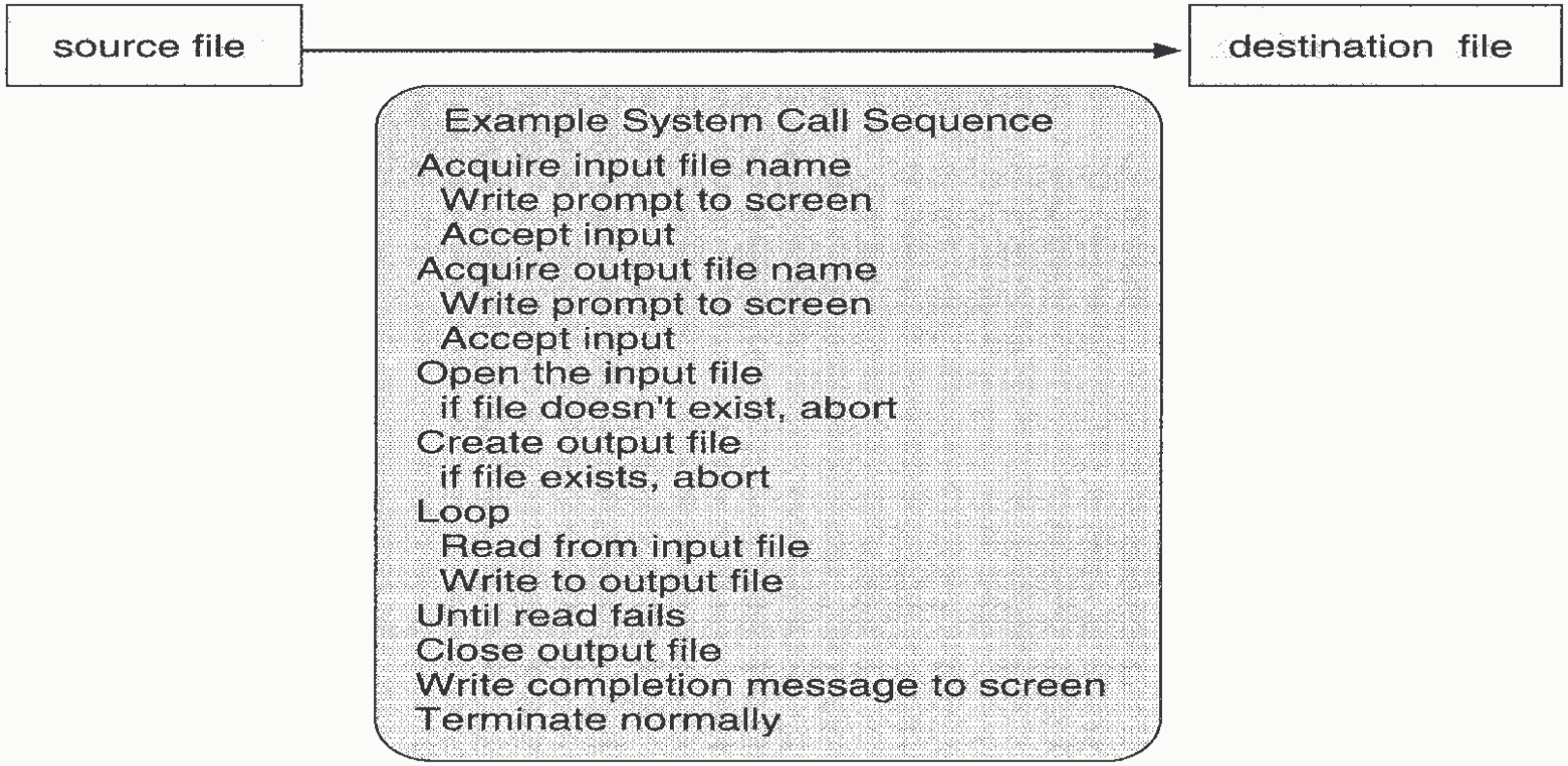
- User interface (UI)
- Program execution
- I/O operation
- File-system manipulation
- Communications
- Error detection
- Resource allocation
- Accounting
- Protection and security

User OS Interface

- User interface (UI)
  - Command-Line Interface (CLI)
  - Batch interface (file)
  - Graphical User Interface (GUI)
- Command Interpreter
  - In the kernel
  - As a special program (e.g., a shell)

System Calls

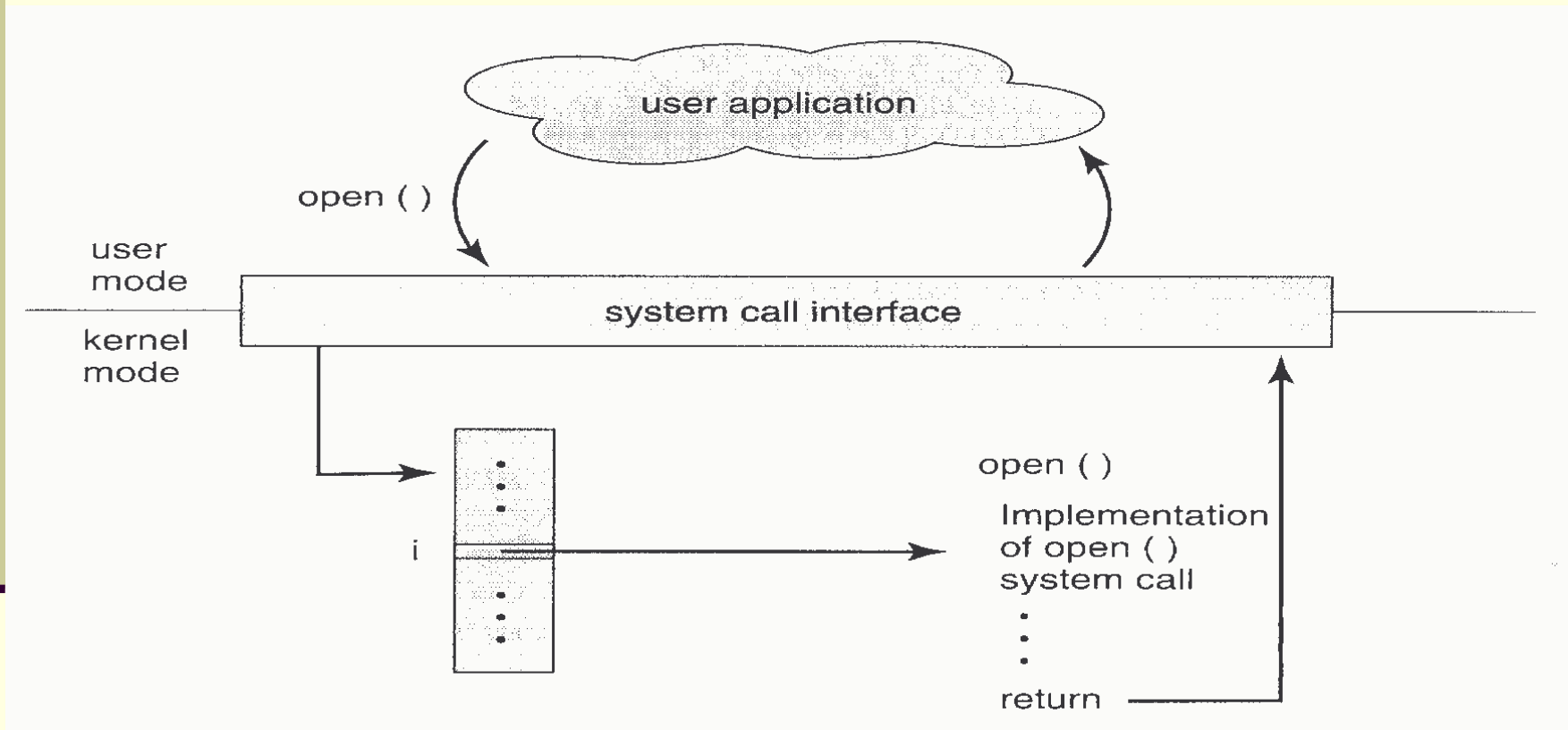
- The basic way to use the OS services
- Each OS has its own system calls



Application Programming Interface (API)

- Standardizing the use of system calls
- Portability of programs on every system supporting the same API (*sending parameters and getting return values*)
  - Win32 API (Windows)
  - POSIX API (UNIX, LINUX, MAC OS X)
  - JAVA API (for programs running on JVM)
- System programs: sequences of system calls to perform more complex operations
  - File management, program loading, …

Handling a System Call



Methods of Passing Parameters

- Registers
- Blocks or tables in memory
- Stack

Types of System Calls

- Process control
- File manipulation
- Device manipulation
- Information maintenance
- Communications
  - Message-passing model: (host name, process name)→ source: client, receiving daemon: server
    - Suitable for transferring smaller amount of data
    - Simpler implementation of communication among computers
  - Shared-memory model:
    - maximum transfer rate
    - Problems: synchronization and protection

Operating System Design and Implementation

- Design Goals
  - At the highest level: batch, time-shared, single-user, ,multi-user, distributed, real-time, …
  - User goals: simple and easy to learn and use, reliable, safe, fast
  - System goals: easy to design, implement, and maintain, flexible, reliable, error free, efficient
  - NO UNIQUE solution for selecting the best requirements among the above

Some definitions

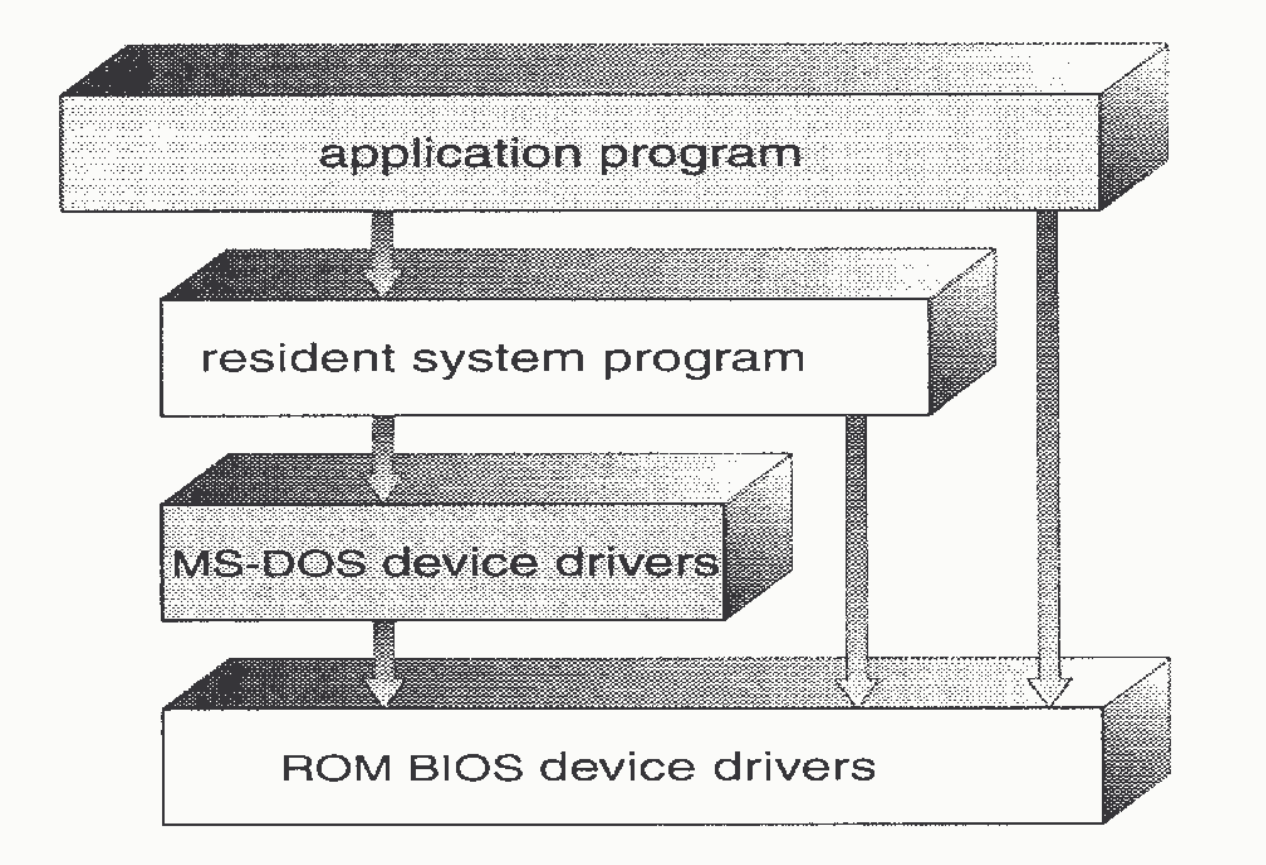
- Mechanism: How to do something
- Policy: What will be done
- Implementation of operating systems
  - Traditionally, written in assembly language
  - Currently, mostly written in C or C++

Operating System Structure

- Simple structure
- Layered approach
- Microkernels (μ-kernels)
- Modules
- Virtual Machines

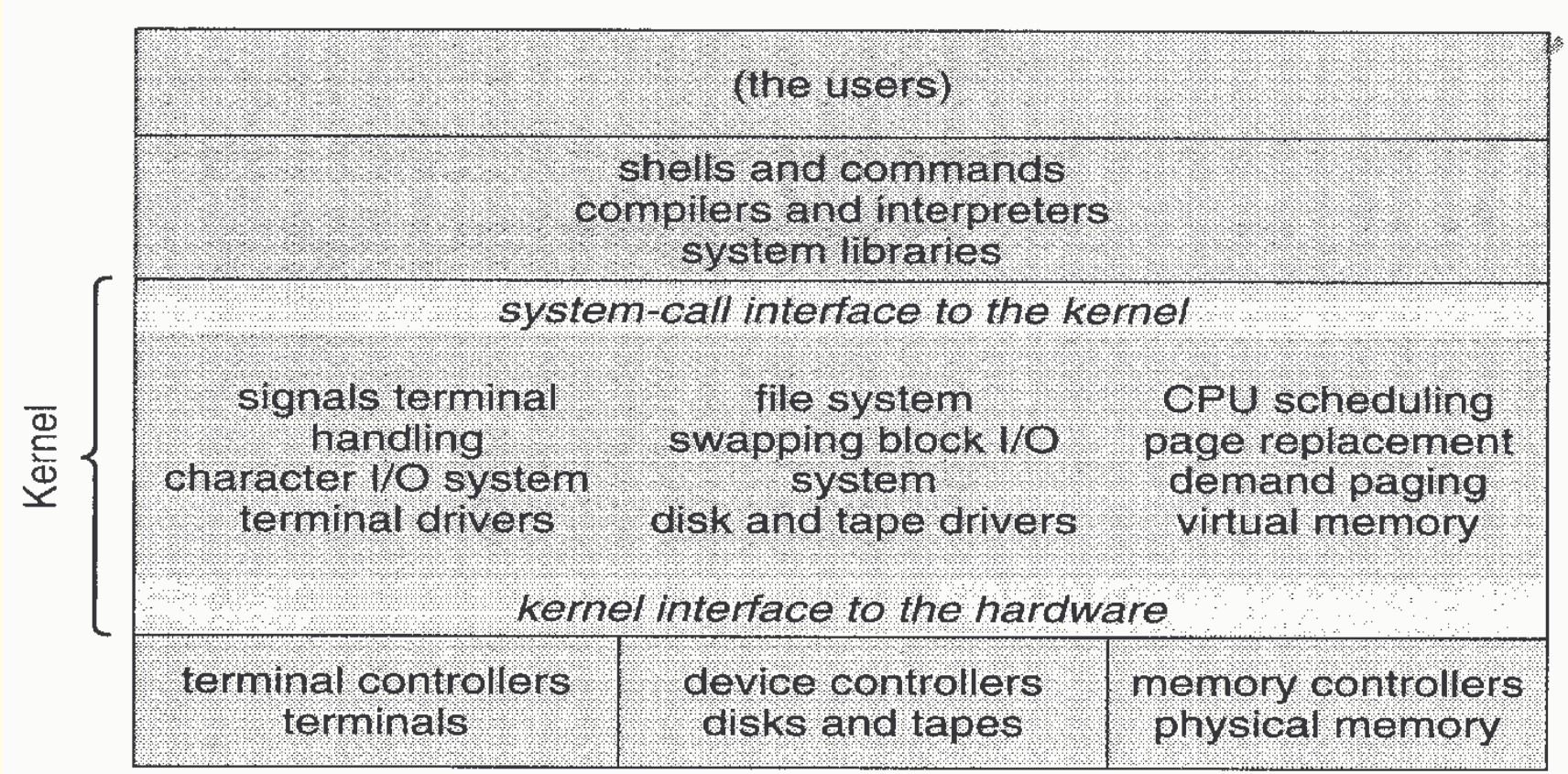
Simple Structure

- MS-DOS layer structure

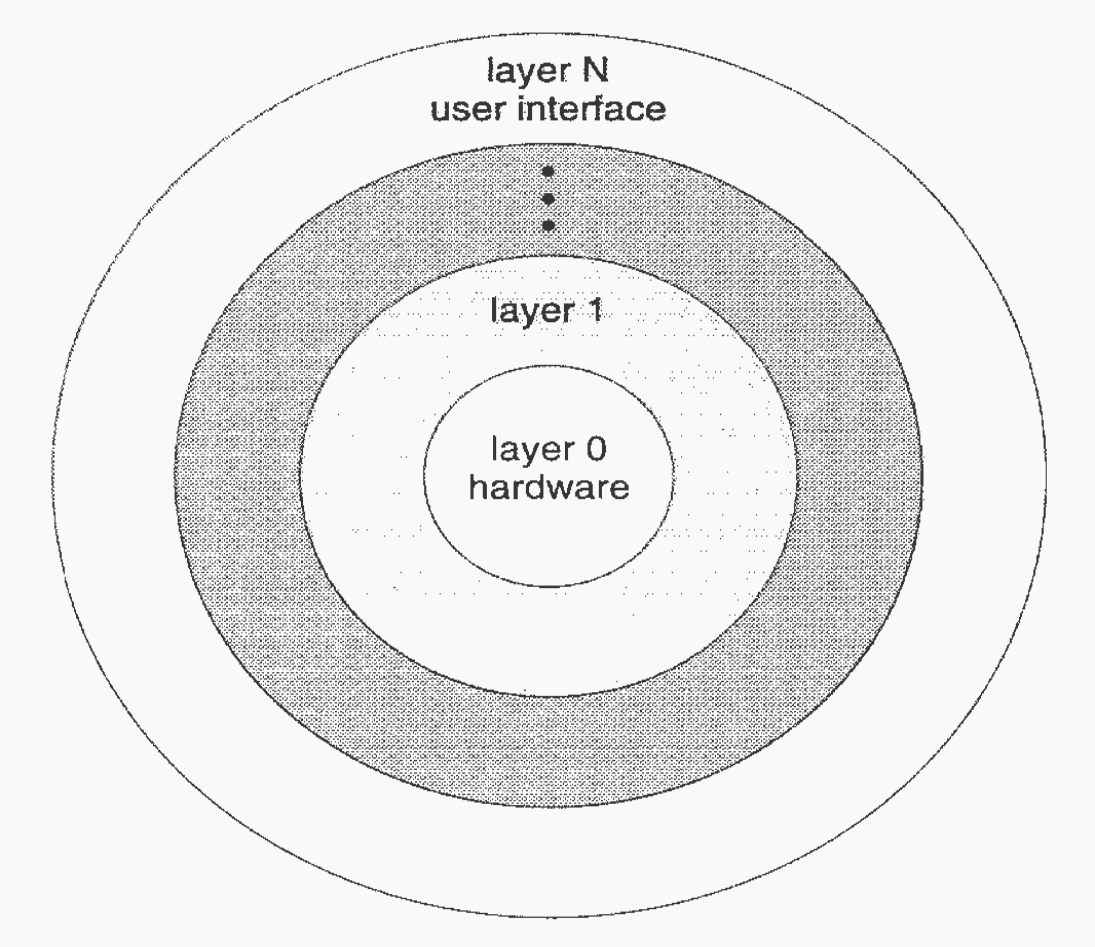


Simple Structure

- UNIX partially-layered structure



Layered Approach



Layered Approach

- Properties
  - Simplicity of construction
  - Simplicity of Debugging
- Problems
  - Precise definition of layers
    - Example: Memory manager requires device driver of backing store (due to virtual memory)
    - The device driver requires CPU scheduler (since if the driver waits for IO, another task should be scheduled)
    - CPU scheduler may require virtual memory for large amount of information of some processes
  - Less efficiency: due to the number of layers a request should pass

Microkernels

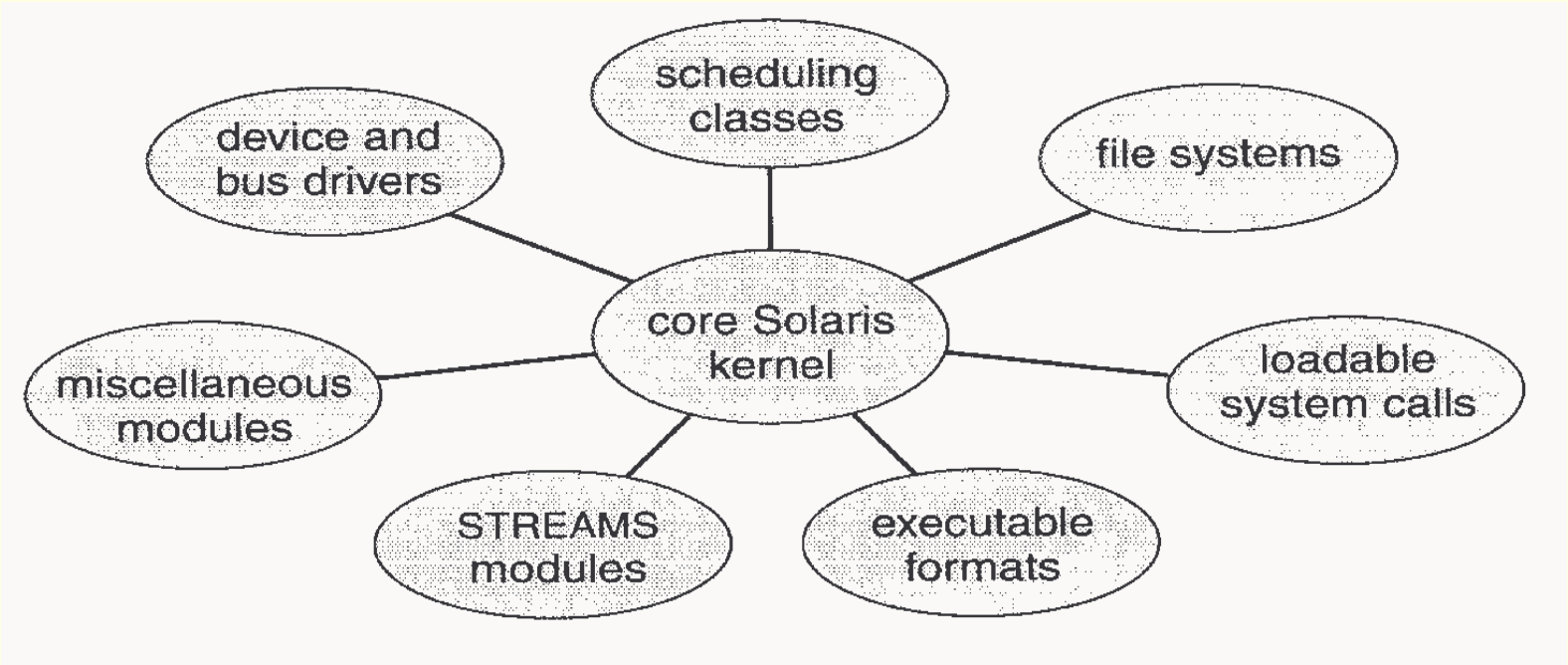
- Removing all nonessential components from the kernel
- Implementing them as system and user-level programs
  - C/S model & the way of finding services
- What μ-kernel does?
  - Minimal process and memory management
  - Communication facilities
    - IPC: Inter-Process Communication
- Advantages
  - Extensibility of the OS
  - Portability
  - Potential for making distributed services
  - More security and reliability (service failure doesn't destroy OS)
- Disadvantages
  - Performance loss (due to IPC)
- Mach, QNX, L4, the first release of Windows NT, …

Modules

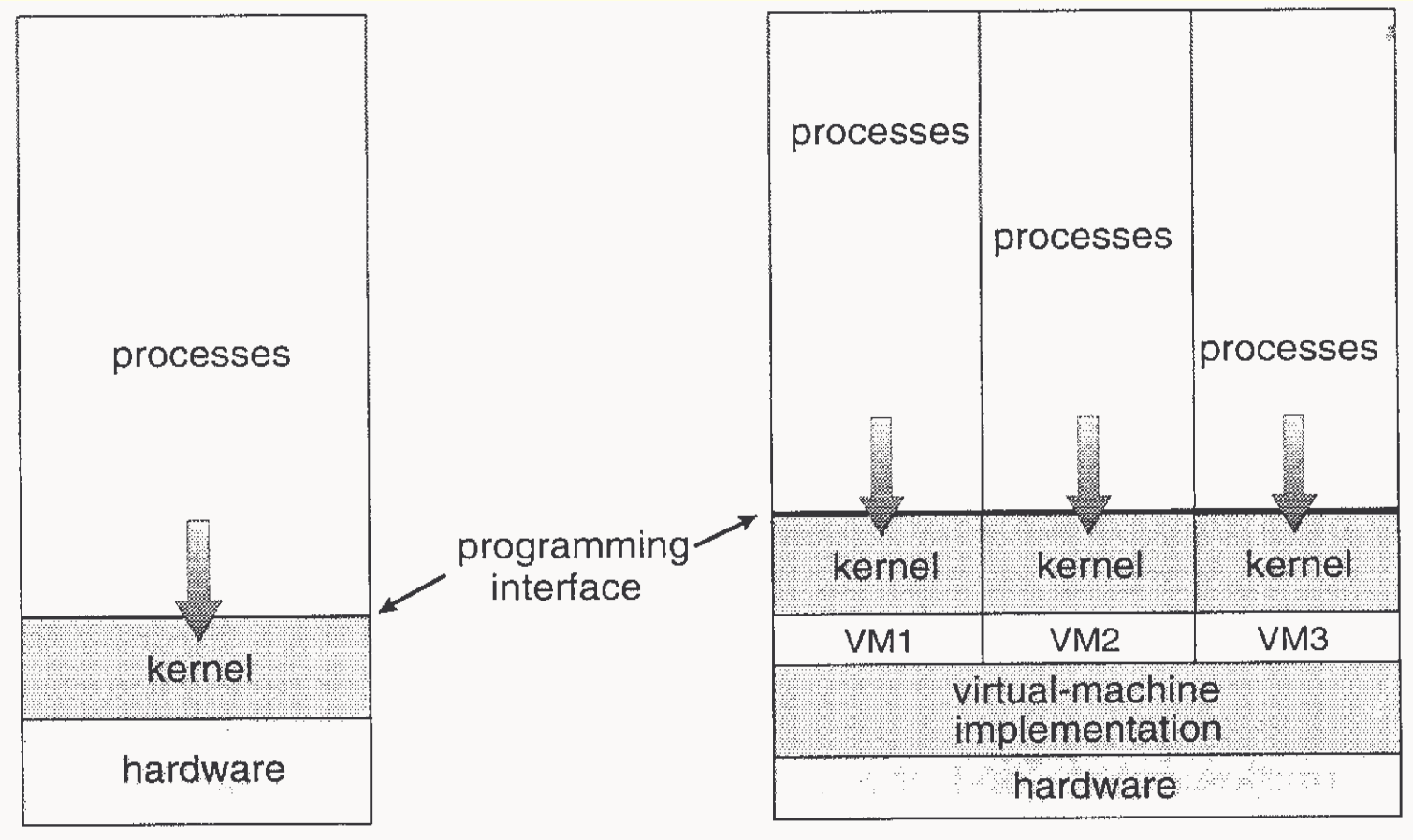
- One of the best current methodologies
  - Using OOP techniques to create a modular kernel
- Kernel: a set of core components
- Dynamically links for additional services (dynamically loadable modules)
  - Boot time or run-time
- Exploits the benefits of both layered and μ-kernel structures:
  - Each module has defined protected interfaces
  - Primary module has only core functions
  - Modules do not need to use message passing to communicate

Modules

- Solaris loadable modules



Virtual Machines



Virtual Machines

- Benefits
  - VM are completely isolated➔ no protection problems
  - No direct sharing of resources
  - Sharing the same HW among different environments with different operating systems running concurrently
  - It is a perfect vehicle for OS research and development
- Main difficulty
  - Disk space➔ solution: minidisks

Virtual Machines

- Implementation
  - User mode
    - Virtual user mode
    - Virtual kernel mode
  - Kernel mode
- The underlying OS may be structured as a layered, μ-kernel, …
- Major difference is time!
  - Real I/O might take 100 ms
  - Virtual I/O might take
    - Less time (because it is spooled)
    - More time (because it is interpreted)